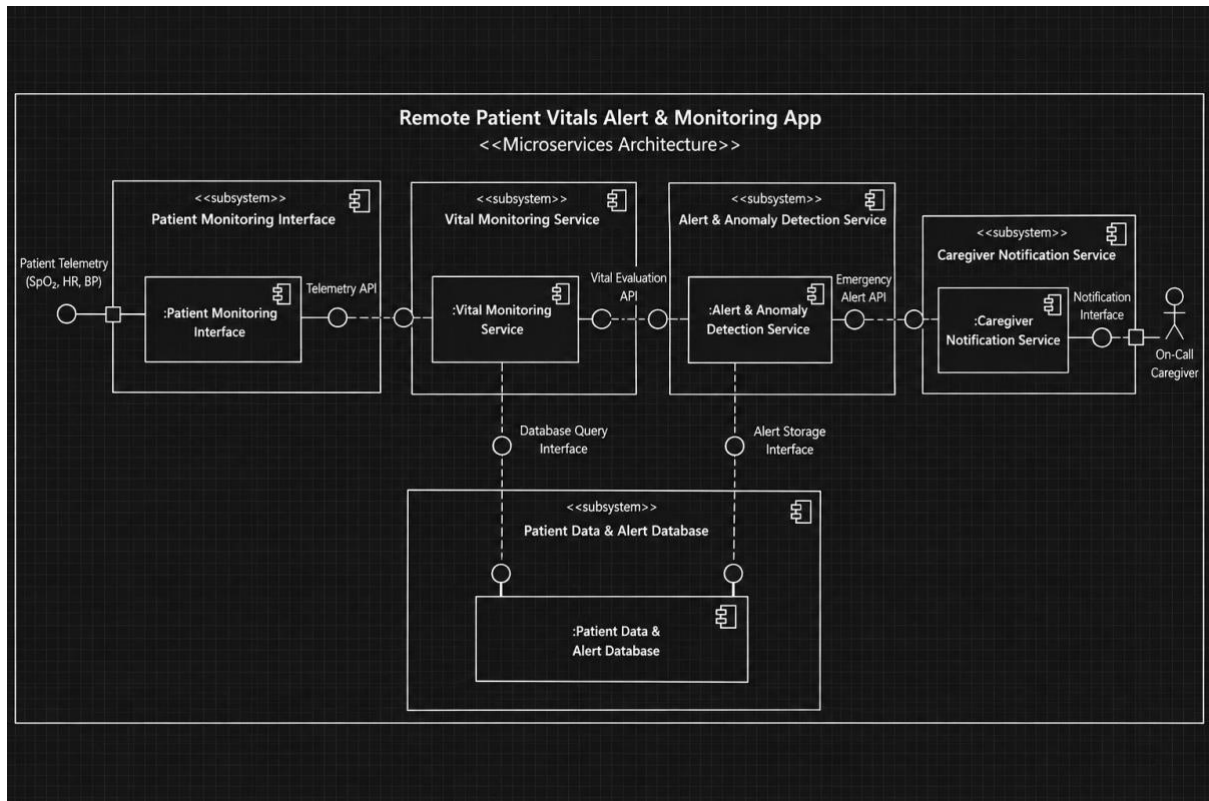


Lab 3: Component Modelling & Architectural Pattern Selection

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Written Justification:

Architecture Selection

We chose Microservices Architecture for the Remote Patient Vitals Alert & Monitoring App.

Architectural Choice and Reasons

1. Independent scalability: The system continuously receives telemetry from multiple patients and must support at least 500 concurrent continuous telemetry streams. With microservices, the Vital Monitoring Service can be scaled independently when telemetry traffic increases, without scaling the entire application.

2. Fault isolation and modularity: The system has separate responsibilities such as vital monitoring, anomaly detection, emergency alert generation, caregiver notification, and data storage. Implementing these as independent services makes the system modular and helps isolate failures. For example, a temporary failure in the notification service does not have to stop the telemetry monitoring service.

Security Advantage

Microservices allow security controls to be applied separately to services handling sensitive patient information. Communication between services can be authenticated and encrypted, while access to patient data can be restricted to authorized components and caregivers. This helps protect patient vital data during transmission and storage.

Performance Benefit

Microservices allow frequently used services, particularly the Vital Monitoring and Alert & Anomaly Detection Services, to be scaled independently. This helps the system process continuous telemetry efficiently and supports the requirement of handling at least 500 concurrent telemetry streams while maintaining reliable monitoring.

Conclusion

Microservices Architecture is therefore appropriate for the Remote Patient Vitals Alert & Monitoring App because it provides independent scalability, modularity, and fault isolation while supporting the system's security and continuous monitoring requirements.